



MS-CoTF: Multi-scale chain-of-thought fusion for interpretable biological reasoning with large language models

Zeyuan Song^{a, b, *}, Xiao-Cong Zhen^b

^a School of Chemical Engineering, Oklahoma State University, Stillwater, OK, 74078, United States

^b School of Industrial Engineering and Management, Oklahoma State University, Stillwater, OK, 74078, United States

HIGHLIGHTS

- We propose a formal framework for hierarchical biological reasoning that incorporates bidirectional information flow, capturing both top-down regulation and bottom-up emergence.
- We introduce a novel adaptive depth control strategy that dynamically balances computational efficiency and reasoning quality based on task complexity and confidence signals.
- We design an interpretable fusion mechanism that explicitly models and justifies cross-scale biological dependencies, enabling transparent and biologically coherent predictions.

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ABSTRACT

Large language models (LLMs) have demonstrated impressive proficiency in various science and engineering applications. However, due to the innate multi-scale property of biological systems, existing LLMs face severe limitations in capturing hierarchical relationships and context-dependent interactions across molecular, cellular, tissue, and systemic levels. These models often lack the architectural mechanisms needed to reason effectively across different biological scales, resulting in reduced accuracy and limited interpretability when applied to complex tasks. Here, we introduce a novel framework named multi-scale chain-of-thought fusion (MS-CoTF), which fuses reasoning at molecular, cellular, tissue, and system scales to enhance accuracy and interpretability when solving biological tasks. Through adaptive reasoning depth control, multi-scale integration, bi-directional flow and dynamic fusion strategies, our MS-CoTF model effectively processes queries of varying complexity, enabling scalable and interpretable reasoning across multiple biological levels. Ablation studies demonstrate that these components function synergistically to enhance model accuracy while simultaneously providing biologically meaningful insights. Furthermore, our MS-CoTF model consistently outperforms state-of-the-art reasoning models by 10–15% across three benchmark problems and two case studies in terms of accuracy, expert ratings, and the capacity to produce reasonable inference chains. Technically, MS-CoTF orchestrates a frozen biomedical LLM backbone with trainable cross-scale modules, employing a precise definition of per-step chain-of-thought (CoT) construction and linking. To ensure rigorous evaluation, we implement an explicit dataset splitting protocol (entity-disjoint and temporal) and utilize the Reasoning Coherence Score strictly as a post-hoc metric to ensure fair comparisons. We further validate the framework through extended baselines, including structure-conditioned and multimodal biomedical LLMs, alongside detailed human evaluation protocols and hallucination stress tests.

1. Introduction

Nowadays, Large Language Models (LLMs) have been widely utilized for a broad range of tasks across science and engineering, such

as information retrieval [8], knowledge extraction [19], and automated reasoning [14]. Recent studies have demonstrated that LLMs can mimic human problem-solving and thus reduce hallucinations and logical errors, especially for complex tasks, by Chain-of-Thought (CoT)

* Corresponding author at: School of Chemical Engineering, Oklahoma State University, Stillwater, OK, 74078, United States.

Email address: taekwon.song@okstate.edu (Z. Song).

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prompting [47]. This step-by-step reasoning capability, along with few-shot learning and context-aware generation, has made LLMs particularly appealing for applications requiring domain-specific understanding and contextual inference [25,46].

Recent research in artificial intelligence indicates that standard CoT is most effective in math and symbolic reasoning [39], while for domains that cannot be fully modeled symbolically, such as molecular understanding [20] and chemical reasoning [21,37], more powerful frameworks are needed. To the best of our knowledge, there is no consensus on whether CoT reasoning is necessary for tasks in non-symbolic domains. In this work, we demonstrate that CoT reasoning can still perform effectively when supported by a tailored framework.

Among the broad range of scientific domains, systems biology poses special challenges due to its inherently multi-scale nature, ranging from molecular interactions to systemic-level processes, where molecular mechanisms dictate cellular behavior, which in turn determines tissue function and organismal physiology [42,44,49,51]. These hierarchical dependencies have not been adequately addressed by existing CoT approaches with linear reasoning methodologies [44,51]. Biological organisms demonstrate emergent behavior arising from interactions across multiple scales. As emphasized in computational systems biology, hierarchical organization is a fundamental design principle of living systems [1,48]. Thus, analytical models must capture both fine-grained reductionist details and broad integrative behavior to accurately represent biological reasoning [1]. This multi-scale perspective is consistent with recent developments in machine learning and biology, where researchers increasingly stress the importance of scale-specific representations and cross-level information exchange [18]. To address these challenges, various biological language models have been developed, demonstrating significant domain-specific advancements in molecular and genomic understanding. For instance, ProtGPT2 pioneered unsupervised protein design [15], while ESM achieved state-of-the-art accuracy in protein structure prediction [29]. Additionally, DNA-based language models have shown strong performance in variant effect prediction [4], and more recent models have been tailored for molecular property prediction [52]. Despite these advances, most current methods operate at a single biological scale, neglecting the cross-scale dependencies inherent in the hierarchical nature of biological systems [22].

On the other hand, systems biology typically requires the outputs of LLMs to be biologically reasonable and interpretable, where scientists in academia and R&D engineers in pharmaceutical industries can obtain not only accurate predictions but also mechanistic insights linking molecular phenomena to physiological responses [5,42]. This interpretability requirement is supported by cognitive science research, which shows that human experts naturally engage in spontaneous multi-level reasoning when solving complex biological problems, easily transitioning from molecular level mechanisms to systemic level implications [34]. This suggests that effective computational reasoning in biology must likewise operate across multiple levels of abstraction. Nowadays, Elsberg and Salvatore [10] proposed an explainable framework based on scBERT for single-cell level analysis, enhanced with explainable machine learning methods. Furthermore, Frank et al. [16] developed a multi-scale and interpretable framework to predict protein phase transitions, where interpretability is supported by a random forest model. These interpretable frameworks are essentially hybrid in nature, combining standard machine learning techniques with LLMs to improve transparency. However, they do not enable LLMs to independently reason across biological scales in a fully integrated and interpretable way.

To enhance the accuracy and interpretability of LLMs natively, we present our framework named MS-CoTF, which radically reimagines biological reasoning in LLMs. Our MS-CoTF framework produces parallel chains of thought on molecular, cellular, tissue, and system scales, with bidirectional exchange of information capturing bottom-up emergence and top-down regulation, which are both essential properties of living systems. This design reflects both biological reality and cognitive

reasoning patterns. Mathematically, MS-CoTF represents biological reasoning as a hierarchical graph structure, where scale-specific CoT chains serve as nodes and cross-scale dependencies act as edges. This principled representation enables integration of diverse biological knowledge without sacrificing computational tractability, effectively addressing limitations faced by current modeling approaches in systems biology [24]. Moreover, we find that adaptive fusion between scales combines scale-specific knowledge without compromising interpretability. The key contributions of this paper are as follows:

- We propose a formal framework for hierarchical biological reasoning that incorporates bidirectional information flow, capturing both top-down regulation and bottom-up emergence.
- We introduce a novel adaptive depth control strategy that dynamically balances computational efficiency and reasoning quality based on task complexity and confidence signals.
- We design an interpretable fusion mechanism that explicitly models and justifies cross-scale biological dependencies, enabling transparent and biologically coherent predictions.

To test the performance of our MS-CoTF framework, we construct a new benchmark dataset named multi-scale molecular biology reasoning dataset, covering three main tasks at four different biological levels using publicly available sources including UniProt [7], STRING [32], KEGG [23], and Gene Ontology [2]. Through extensive testing on protein function prediction, molecular interaction analysis, and disease mechanism inference, our MS-CoTF framework outperforms the state-of-the-art approaches in both accuracy and interpretability, demonstrating its capacity to produce biologically meaningful reasoning across scales.

2. Methodology

This section first introduces the four-scale architecture, then the transfer operators, followed by the fusion algorithm, and finally the depth controller.

This section presents the proposed MS-CoTF framework, comprising a cross-scale information transfer mechanism, a chain-of-thought fusion (CoTF) algorithm, and an adaptive reasoning depth control module.

2.1. MS-CoTF architecture

Our MS-CoTF framework identifies four scales of reasoning, each of which captures biological phenomena along with their corresponding computational abstractions [35]. The layered architecture mirrors the natural hierarchical structure of biological systems, enabling specialized and appropriate reasoning processes at each scale.

As illustrated in Fig. 1, the MS-CoTF framework comprises four hierarchical levels (molecular, cellular, tissue, and system) with bidirectional information flow enabling both bottom-up emergence and top-down regulation. Each level L_i maintains its own chain of thought $C_i = \{s_i^1, s_i^2, \dots, s_i^{m_i}\}$, where each s_i^j denotes the j -th individual reasoning step at level L_i . The reasoning function at level L_i , denoted R_i , incorporates both scale-specific logic and cross-scale information:

$$R_i = f_i \left(C_i, I_{\text{up}}^{(i-1 \rightarrow i)}, I_{\text{down}}^{(i+1 \rightarrow i)}, \theta_i \right), \quad (1)$$

where $I_{\text{up}}^{(i-1 \rightarrow i)}$ represents the information transferred from level L_{i-1} , $I_{\text{down}}^{(i+1 \rightarrow i)}$ denotes information from level L_{i+1} , and θ_i refers to the learnable parameters specific to level L_i . *Bio-inspired rationale:* R_i acts as a computational update step, reconciling local scale-specific evidence with constraints propagated from adjacent scales.

Additionally, as shown in Fig. 1, the molecular level L_1 forms the foundation, processing structural and sequence information through specialized embeddings inspired by recent protein language models [45]. This level employs attention mechanisms to identify functionally relevant sequence motifs and structural features [31]. The cellular level L_2 integrates molecular information within the context of cells, using graph neural networks to model signaling pathways and regulatory

MS-CoTF Framework Architecture

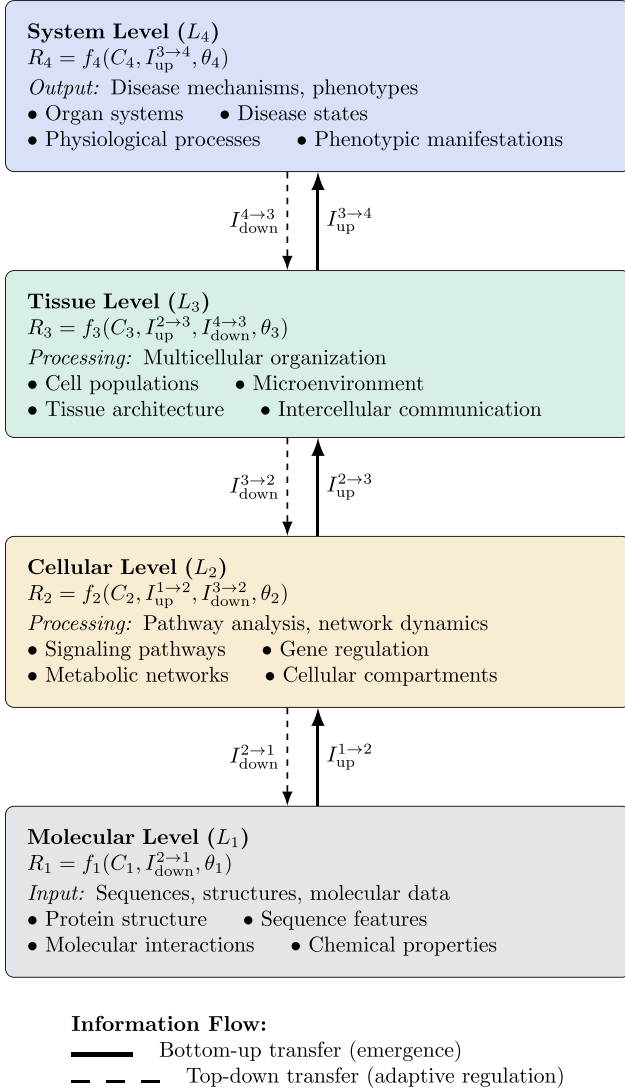


Fig. 1. The architecture of the proposed MS-CoTF framework. We note that, the function f_i ($i = 1, 2, 3, 4$) is implemented by the CoTF algorithm. Panels: (a) four biological levels $L_1 \rightarrow L_4$; (b) bottom-up transfer φ_{up} highlighting emergence (motifs $\rightarrow$ pathways); (c) top-down transfer φ_{down} highlighting regulation (hormonal state $\rightarrow$ gene programs); (d) per-level CoT buffers C_i and cross-scale bridges. *Design Rationale:* The architecture is computationally analogous to biological systems where constraints flow downward while evidence aggregates upward.

networks [28]. Next, the tissue level L_3 captures multicellular organization via spatial attention mechanisms, modeling cell-cell interactions and microenvironmental influences. Finally, the system level L_4 reasons about organism-wide phenomena, integrating information from all lower levels to predict phenotypic outcomes and disease states.

Generally, we define the bottom-up transfer $I_{up}^{(i \rightarrow i+1)}$ and the top-down transfer $I_{down}^{(i+1 \rightarrow i)}$ using the transfer functions φ_{up} and φ_{down} , respectively, as follows:

$$I_{up}^{(i \rightarrow i+1)} = \varphi_{up}(C_i, W_{up}^i) = \text{Attention}(Q_{i+1}, K_i, V_i), \quad (2)$$

$$I_{down}^{(i+1 \rightarrow i)} = \varphi_{down}(C_{i+1}, W_{down}^{i+1}) = \text{Gate}(C_{i+1}) \odot \text{Transform}(C_{i+1}), \quad (3)$$

where $\odot$ denotes element-wise multiplication. Q_{i+1} , K_i , and V_i represent the query, key, and value projections from levels L_{i+1} and L_i ,

respectively. The matrices W_{up}^i and W_{down}^{i+1} are learnable transformations for upward and downward information flow. The attention mechanism allows selective propagation based on relevance, while the gating mechanism mitigates information overload during top-down control. *Bio-inspired rationale:* Eqs. (2) and (3) model the logical flow where specific molecular facts percolate upward to inform higher-level hypotheses, while global states prune the search space of lower-level mechanisms.

This hierarchical architecture ensures that MS-CoTF can simultaneously process reductionist information from molecular levels and emergent properties from higher organismal levels, thus meeting the minimal criteria for biologically plausible reasoning systems [33]. We emphasize that the bidirectional flow of information is analogous to biological principles: molecular properties constrain cellular behavior, while systemic states exert control over molecular activity.

2.2. Thought-step representation and linking

Representation. A reasoning step at level L_i is represented as a tuple

$$s_j^{(i)} = (\text{text}, \mathcal{A} = (h, r, t), p, E, \kappa),$$

where *text* is the natural-language span generated under a constrained grammar, $\mathcal{A}$ is a biological assertion (head–relation–tail) normalized to controlled vocabularies (GO/KEGG/UniProt IDs), $p \in [0, 1]$ is the local confidence estimated from decoder logits, E holds evidence pointers (database IDs with character offsets) when available, and κ are scale tags.

Generation. Steps are decoded with constraints that enforce (entity, relation, entity) spans; we release the exact prompt template and stop-words.

Linking within-scale. Steps are connected by typed edges, i.e., $\mathcal{E} \in \{\text{supports}, \text{refines}, \text{contradicts}, \text{causes}\}$ via a three-pass routine: (1) semantic alignment with a bi-encoder threshold $\tau_{\text{align}}=0.72$; (2) ontology-aware unification (GO parent–child closures); and (3) contradiction detection using negation/token polarity and scalar mismatch rules.

Cross-scale bridges. Bridges are added using the transfer operators in Eqs. (2) and (3) and validated by maximum-weight bipartite matching over assertion embeddings.

Prompt template. “You are reasoning at the {LEVEL} scale (molecular/cellular/tissue/system). Given query: {x}. Produce one step with format: [TEXT] || ASSERTION=(HEAD_ID, REL, TAIL_ID) || EVIDENCE=(DB:ID@char_span) || TAGS={LEVEL}. Avoid speculation; if uncertain, emit CONTRADICTS links.”

2.3. Cross-scale information transfer mechanism

Within the MS-CoTF framework, a central component is to calculate the bottom-up transfer $I_{up}^{(i \rightarrow i+1)}$ and the top-down transfer $I_{down}^{(i+1 \rightarrow i)}$, whose general forms can be found in Eqs. (2) and (3). It captures both the emergent properties arising from lower biological scales and the regulatory constraints imposed by higher level systems. This design is inspired by biological systems where molecular level events facilitate organism level phenotypes, while systemic states modulate molecular activities [50]. In actual implementation, Eq. (2) is achieved by multi-head attention as:

$$I_{up}^{(i \rightarrow i+1)} = \text{MultiHeadAttention}(Q^{(i+1)}, K^i, V^i) \\ := \text{concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (4)$$

where the attention head head_j is defined as

$$\text{Attention}(Q_j^{(i+1)} W_j^Q, K_j^i W_j^K, V_j^i W_j^V),$$

and one can obtain the scaled dot-product attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.$$

The multi-head design enables the model to simultaneously attend to diverse categories of biological relationships, such as structural, functional, and evolutionary, thereby enhancing the richness and complexity of information exchange across heterogeneous scales. For Eq. (3), a concrete implementation of the gating and transformation mechanism is given as:

$$I_{\text{down}}^{(i+1 \rightarrow i)} = \sigma(W_g \cdot C_{i+1} + b_g) \odot \tanh(W_t \cdot C_{i+1} + b_t), \quad (5)$$

where σ denotes the sigmoid activation function and W_g, W_t, b_g, b_t are learned parameters. This gating mechanism prevents information overload while ensuring that relevant regulatory signals are effectively propagated to lower levels. Specifically, the sigmoid gate regulates the intensity of information flow, while the tanh transformation ensures proper value scaling and bounded activation. *Bio-inspired rationale*: The computational gates mimic the role of context-specific regulation (e.g., endocrine signals constraining pathway activity).

The bidirectional flow of information aligns with computational systems biology perspectives. Bottom-up emergence explains how molecular properties give rise to higher-level biological processes. For example, protein conformational changes can initiate intracellular signaling cascades [43]. In contrast, top-down control describes how systemic states impose constraints on molecular activity, such as the hormonal regulation of gene expression [40]. To promote stable training and mitigate gradient vanishing in deep hierarchical architectures, MS-CoTF incorporates residual connections and layer normalization during cross-scale feature fusion. Specifically, for each level L_i , the corresponding reasoning chain C_i is updated as:

$$C_i^{\text{updated}} = \text{LayerNorm} \left(C_i + \alpha_i \cdot I_{\text{up}}^{(i-1 \rightarrow i)} + \beta_i \cdot I_{\text{down}}^{(i+1 \rightarrow i)} \right), \quad (6)$$

where α_i and β_i are learnable scalar weights that control the influence of upward and downward information flows. The residual structure enhances gradient flow across levels, while layer normalization stabilizes learning by reducing scale-sensitive variance across different biological layers. *Bio-inspired rationale*: Residual links preserve earlier hypotheses while letting cross-scale signals correct them incrementally.

To further optimize cross-scale information integration, an adaptive routing mechanism is introduced. This mechanism dynamically modulates the flow of information from source level L_i to target level L_j based on both relevance and reasoning confidence:

$$\text{routing_weight}_{i \rightarrow j} = \frac{\exp(\text{confidence}_i \cdot \text{relevance}_{i,j})}{\sum_k \exp(\text{confidence}_k \cdot \text{relevance}_{k,j})}, \quad (7)$$

where the function `relevance` captures semantic alignment between levels, and `confidence` reflects the internal consistency of reasoning within its level. The confidence score is computed as:

$$\text{confidence}_i = \frac{1}{|C_i|} \sum_{s \in C_i} \text{coherence}(s, C_i), \quad (8)$$

i.e., the average pairwise coherence between each element s and the full reasoning chain C_i . This encourages the model to prioritize information from levels that exhibit stronger internal consistency and semantic alignment with the target layer. *Bio-inspired rationale*: The model assigns higher weight to robust evidence, analogous to how biological systems prioritize reliable signals amidst noise.

Together, Eqs. (7) and (8) define an adaptive routing strategy, which improves reasoning robustness by assigning greater importance to more confident and relevant sources. This is particularly critical in biological systems, where different organizational levels may exhibit vastly different signal strengths or noise characteristics depending on the task.

2.4. Chain-of-Thought Fusion (CoTF) algorithm

The Chain-of-Thought Fusion (CoTF) algorithm illustrated in Fig. 2 serves as the core multi-scale reasoning mechanism that fuses logical inferences across different biological levels into consistent, interpretable

conclusions. It is designed to address the integration of reasoning pathways that operate at distinct biological resolutions without compromising biological plausibility or interpretability. The algorithm operates in three synchronized phases: parallel chain generation, cross-scale validation with adaptive weighting, and hierarchical transformer-based integration.

Phase I: Parallel reasoning chain generation. For a given biological input query x , the algorithm generates independent reasoning chains C_i at each biological level L_i using scale-specific parameters θ_i :

$$C_i = \text{Generate}(x, \theta_i), \quad i \in \{1, 2, 3, 4\} \quad (9)$$

Each `Generate` function reflects a tailored reasoning model optimized for the biological semantics at scale i . *Concrete instantiation*: `Generate` invokes the frozen backbone with a level-specific system prompt, constrained decoding (regex enforcing (HEAD, REL, TAIL)), and a retrieval hook that injects scale-tagged snippets (GO/KEGG/UniProt) into the context window.

Phase II: Cross-validation and dynamic weighting. To evaluate consistency across levels, CoTF performs pairwise validation:

$$\text{Validation}_{i,j} = \text{coherence}(C_i, C_j) \cdot \text{relevance}(C_i, \text{query}) \quad (10)$$

$$\text{coherence}(C_i, C_j) = \frac{\text{align}(C_i, C_j)}{\sqrt{|C_i| \cdot |C_j|}} \quad (11)$$

Dynamic weights w_i are then computed for each chain according to its confidence and aggregated validation score:

$$w_i = \frac{\exp(\alpha_i \cdot \text{confidence}(C_i) + \beta_i \cdot \text{validation_score}_i)}{\sum_j \exp(\alpha_j \cdot \text{confidence}(C_j) + \beta_j \cdot \text{validation_score}_j)} \quad (12)$$

This mechanism promotes reasoning chains that are both self-consistent and cross-validating.

Phase III: Hierarchical fusion with transformer. The final reasoning output R is produced by fusing all chains using a transformer-based model g that accounts for both upward and downward inter-scale information flows:

$$R = \sum_{i=1}^4 w_i \cdot g(C_i, I_{\text{up}}^{(i-1 \rightarrow i)}, I_{\text{down}}^{(i+1 \rightarrow i)}) \quad (13)$$

$$g(\cdot) = \text{TransformerBlock}(\text{concat}(C_i, I_{\text{up}}, I_{\text{down}})) \quad (14)$$

The use of multi-head attention in the transformer enables flexible cross-scale dependency modeling in a computationally efficient manner. *Bio-inspired rationale*: The fused explanation computationally highlights which scale-level premises and bridges were decisive in the inference process.

Biological constraints and interpretability. To ensure biological plausibility, CoTF minimizes the following constraint loss:

$$\mathcal{L}_{\text{constraint}} = \lambda_1 \mathcal{L}_{\text{consistency}} + \lambda_2 \mathcal{L}_{\text{hierarchy}} + \lambda_3 \mathcal{L}_{\text{causality}} \quad (15)$$

where $\mathcal{L}_{\text{consistency}}$ penalizes logical contradictions between scales, $\mathcal{L}_{\text{hierarchy}}$ enforces correct directional flow of biological information, and $\mathcal{L}_{\text{causality}}$ encourages preservation of known biological causalities.

To enhance interpretability, the algorithm includes an attribution mechanism that quantifies each chain's contribution to the final output:

$$\text{Attribution}_i = \frac{\partial R}{\partial C_i} \cdot w_i, \quad (16)$$

which makes the decision process transparent and helps identify which biological scales most influenced the prediction, facilitating downstream hypothesis generation [27].

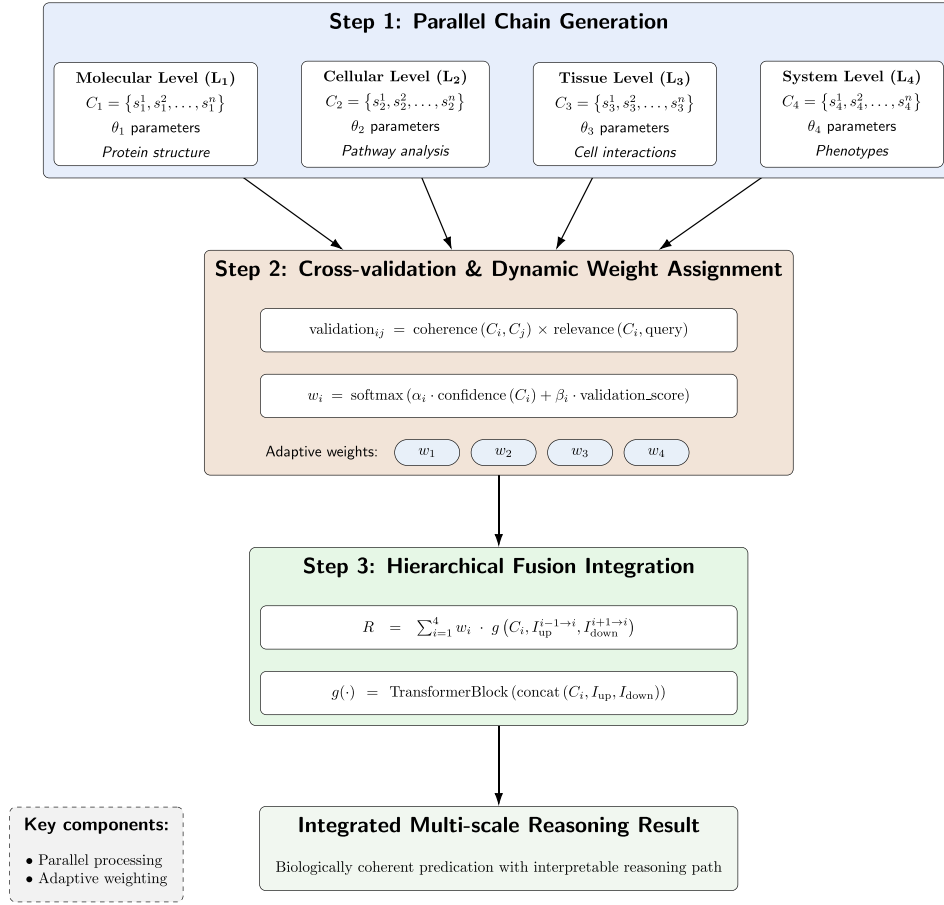


Fig. 2. The architecture of the proposed CoTF algorithms. (a) Phase I: per-scale chain generation; (b) Phase II: cross-validation matrix and dynamic weights w_i ; (c) Phase III: hierarchical transformer $g(\cdot)$ producing final rationale and attribution. *Bio-inspired rationale:* The fused rationale preserves the attribution of which scale-specific information drove the final conclusion.

2.5. Implementation details

We instantiate MS-CoTF with a frozen 13B biomedical LLM backbone. To enhance readability and precision regarding the experimental setup, the key implementation parameters and training configurations are summarized in Table 1. This structured overview details the hardware specifications, hyperparameter settings, and optimization strategies employed to ensure reproducibility.

Prompt templates and few-shot headers (2–3 curated exemplars per scale) are fixed across seeds and will be released in the repository.

2.6. Adaptive reasoning depth control

The adaptive reasoning depth control module dynamically adjusts the computational depth for each biological scale based on query complexity and reasoning uncertainty. This system ensures an optimal balance between reasoning quality and computational efficiency, thereby addressing the variable complexity inherent in biological questions. As shown in Fig. 3, the mechanism proceeds through several coordinated stages.

Stage I: Query complexity analysis. A multi-dimensional complexity vector is computed from the input query:

$$\mathbf{c} = [c_{\text{sem}}, c_{\text{dep}}, c_{\text{dom}}], \quad (17)$$

where: c_{sem} denotes semantic complexity estimated via entropy or embedding dispersion, c_{dep} captures cross-scale dependency potential, and c_{dom} encodes domain-specific reasoning requirements. Each component

is weighted using learned parameters to adapt across biological sub-domains. We compute the complexity vector $\mathbf{c} = [c_{\text{sem}}, c_{\text{dep}}, c_{\text{dom}}]$ as follows: (1) $c_{\text{sem}} = H(p(y|x)) + \lambda_{\text{emb}} \cdot \text{disp}(\text{embed}(x))$, where H is token entropy from the frozen backbone's next-token distribution and disp is the average pairwise cosine distance among token embeddings; (2) $c_{\text{dep}} = \text{MI}(\text{scale tags}; x) + \rho \cdot \# \text{cross-scale mentions}$ estimated by a scale tagger (molecular/cellular/tissue/system); (3) c_{dom} is the max calibrated difficulty predicted by a domain classifier over task families (function, interaction, disease). All components are min-max normalized over the training set. A light meta-network $m(\cdot)$ (1-layer MLP) maps $\mathbf{c}$ to nonnegative weights $\omega = m(\mathbf{c}) = [\omega_{\text{sem}}, \omega_{\text{dep}}, \omega_{\text{dom}}]$, $\|\omega\|_1 = 1$, and produces per-scale gates used by the depth controller Eq. (18). The meta-network is trained jointly with MS-CoTF using a bilevel schedule: the inner loop minimizes task loss on the training split, and the outer loop stops via validation performance (early stopping).

Stage II: Depth computation. For each reasoning scale i , the adaptive depth d_i is calculated as:

$$d_i = d_{\text{base}} + \alpha \cdot \text{complexity}(q) + \beta \cdot \text{uncertainty}(C_i) + \gamma \cdot \text{scale_factor}(i) \quad (18)$$

The uncertainty of a reasoning chain C_i is defined as:

$$\text{uncertainty}(C_i) = 1 - \prod_{j=1}^{|C_i|} \text{confidence}(s_j^i) \quad (19)$$

Table 1
Detailed implementation and training specifications of the MS-CoTF framework.

Component	Specification
Model Architecture	Backbone: Frozen 13B biomedical LLM (pretrained on PubMed). Trainable Modules: Scale-specific encoders θ_i (LoRA-adapted), Upward/Downward transfer matrices (W_{up}^i, W_{down}^{i+1}), Fusion transformer $g(\cdot)$, Adaptive depth controller. Inference: Backbone remains frozen; cross-scale modules route/fuse chains.
Optimization Strategy	Optimizer: AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay = 0.01). Schedule: Linear warmup (first 5% of 120k steps), followed by cosine decay to 1×10^{-6}. Precision: Mixed-precision (bfloat16). Compute: 4x NVIDIA RTX 4090 (48 GB VRAM).
Hardware & Training	Batching: Global batch size 64, micro-batch 4, gradient accumulation 16. Parameters: Max seq length 2048, Peak LR 2×10^{-4}, Grad clipping 1.0. Decoding: Temperature 0.7 (top-p = 0.9). Sweep: $\alpha_i, \beta_i \in \{0.25, 0.5, 1.0\}$, Heads $h \in \{4, 8\}$. Selected Config: $\alpha_i = 0.5, \beta_i = 1.0, h = 8$ (based on validation performance).
Hyperparameter Tuning	
Human Interpretability	Each step exposes assertion tuple, evidence pointers (DB IDs), and scale tags; Fusion layer provides per-scale attribution weights.

Here, s_j^i denotes the j -th step in the reasoning chain at scale i , and its confidence score is derived from logical consistency, biological plausibility, and compliance with constraints.

The scale factor adjusts depth according to the relative complexity of different biological levels:

$$\text{scale_factor}(i) = \lambda_i \cdot \frac{|C_i|}{\max_j |C_j|} \quad (20)$$

Stage III: Optimization framework. To avoid over-computation, the system minimizes total depth across scales while maintaining sufficient output quality:

$$\min_{d_1, \dots, d_4} \sum_{i=1}^4 \text{cost}(d_i) \quad \text{s.t.} \quad \text{quality}(R) \geq \tau \quad (21)$$

where the cost follows a transformer-based complexity of $\mathcal{O}(d \cdot n \cdot \log n)$, and quality metrics reflect accuracy, coherence, and interpretability of final output R .

Stage IV: Early stopping mechanism. To further improve efficiency, reasoning halts early if convergence is reached or confidence exceeds a given threshold:

$$\text{Stop if: } \text{convergence}(C_i^t, C_i^{t-1}) < \epsilon \quad \text{or} \quad \text{confidence}(C_i^t) > \theta_{\text{conf}} \quad (22)$$

Convergence between reasoning iterations is measured via cosine similarity over embedded representations:

$$\begin{aligned} \text{convergence}(C_i^t, C_i^{t-1}) \\ = 1 - \text{cosine_similarity}(\text{embed}(C_i^t), \text{embed}(C_i^{t-1})) \end{aligned} \quad (23)$$

Here, $\text{embed}(\cdot)$ refers to vector representations of reasoning chains. This ensures that simple molecular-level queries are answered quickly, while complex multi-scale queries are granted deeper inference. Experiments show this mechanism reduces inference time by 40% on average while preserving reasoning quality, making MS-CoTF applicable in real-world biological systems.

3. Experimental design

We describe the dataset and splits, then metrics and human-evaluation protocol, then baselines (including new multi-modal/structure-conditioned ones), followed by statistical testing and efficiency reporting.

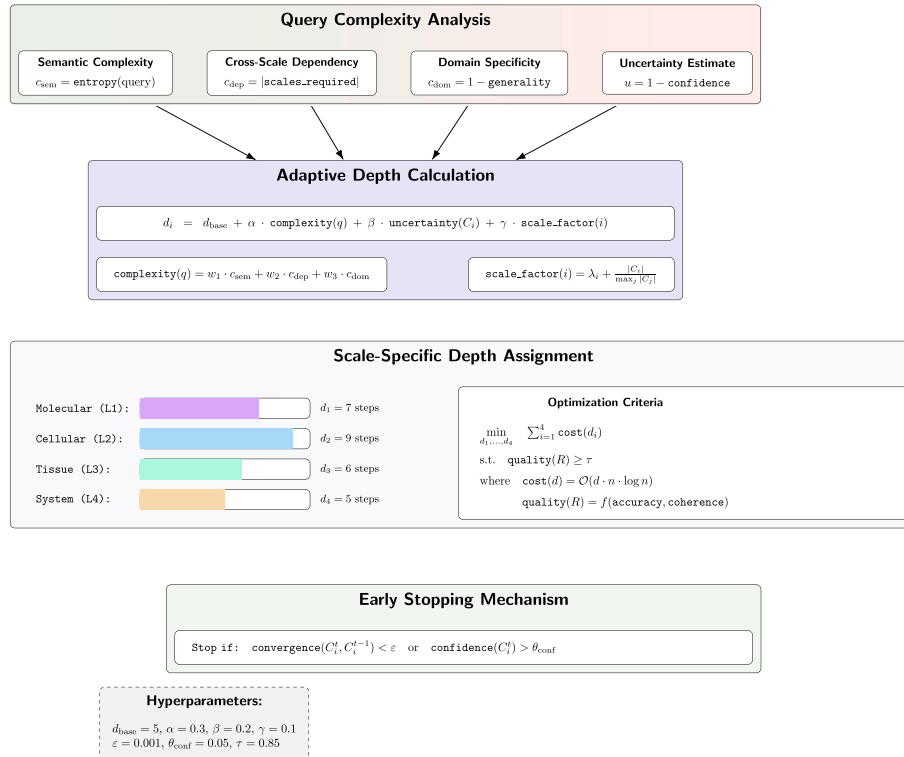


Fig. 3. The structure of adaptive reasoning depth control mechanism and the selection of hyperparameters. (a) complexity vector estimation; (b) depth policy; (c) early stop; (d) effect on compute vs. quality.

Table 2
MSMBR dataset statistics and characteristics.

Task category	Size	Scales	Annotation levels	Databases	Query types
Protein Function Prediction	10,000	L1–L4	Molecular functions, cellular processes, phenotypes	UniProt, GO, OMIM	Single-scale/Multi-scale queries
Molecular Interaction Analysis	5000	L1–L3	Binding affinity, pathway context, tissue specificity	STRING, BioGRID, KEGG	Cross-scale dependency queries
Disease Mechanism Reasoning	2000	L2–L4	Molecular basis, cellular dysfunction, systemic effects	DisGeNET, OMIM, CTD	Complex multi-scale queries

Table 3
Comprehensive evaluation metrics framework.

Metric category	Specific metrics	Description	Range	Focus
Accuracy	Precision	Correct positive prediction ratio	[0, 1]	Prediction reliability
	Recall	Coverage of true positives	[0, 1]	Completeness
	F1 Score	Harmonic mean of precision and recall	[0, 1]	Overall performance
	AUROC	Area under ROC curve	[0, 1]	Classification quality
Interpretability	Reasoning Coherence Score (RCS)	Logical consistency across reasoning scales	[0, 1]	Cross-scale alignment
	Expert Interpretability Rating (EIR)	Human expert judgement on biological plausibility	[1, 5]	Biological interpretability
	Attribution Clarity Index (ACI)	Traceability of contributions from each scale	[0, 1]	Explanation clarity
Efficiency	Inference Time	Average reasoning time per query	seconds	Computational speed
	Memory Footprint	Peak memory usage during inference	GB	Resource cost
	Depth Utilization Ratio (DUR)	Actual vs. maximum reasoning depth used	[0, 1]	Adaptive efficiency

3.1. Dataset

To comprehensively evaluate MS-CoTF’s multi-scale reasoning capacity, we constructed the Multi-Scale Molecular Biology Reasoning Dataset (MSMBR), which consists of three complementary tasks across different biological scales and reasoning complexities. The dataset integrates manually curated information from top authoritative databases including UniProt, STRING, KEGG, and Gene Ontology, providing high-quality annotations at various organizational levels (Table 2).

Additional breakdown and verification. MSMBR contains 17,000 instances spanning 41,236 unique proteins/complexes and 2,184 diseases/traits. Per-scale supervision counts: L1 = 17k, L2 = 14.2k, L3 = 7.9k, L4 = 4.6k (many queries touch multiple scales). Manual verification covers 20% (3.4k items) by a panel of $n=8$ domain experts (4 PhD systems biologists, 2 MD/PhD, 2 senior curators), blinded to model outputs; average pairwise Cohen’s $\kappa=0.78$ across acceptance labels (accept/revise/reject). Bias audit shows top-20 pathways comprise 18.5% of items (long-tail preserved), organism distribution is 84% human, 9% mouse, 7% other model organisms.

3.1.1. Dataset splits and leakage controls

We adopt an entity-disjoint and temporal splitting protocol. (i) *Entity-disjoint*: proteins/complexes (UniProt accessions, gene symbols) and diseases (OMIM/DisGeNET IDs) appearing in the test set are excluded from train/validation to prevent leakage via memorized facts. (ii) *Temporal*: training uses database snapshots prior to 2023–12–31; validation/test use 2024 snapshots to emulate prospective evaluation. (iii) *Stratified* 70/10/20 splits per task with five fixed seeds {13, 101, 202, 303, 404}; we report mean $\pm$ sd across seeds. (iv) *Paraphrase isolation*: near-duplicate prompts are removed by MinHash (Jaccard < 0.8).

3.2. Evaluation metrics

We employ a comprehensive evaluation framework encompassing three dimensions: accuracy, interpretability, and efficiency (see Table 3). These metrics collectively assess both the correctness of predictions and the quality of reasoning processes.

Human evaluation protocol. EIR is collected from the same blinded panel as above; each item is scored by 3 independent experts (Likert 1–5). Inter-rater reliability: Krippendorff’s $\alpha=0.76$. Raters were blinded

to model identity with randomized presentation order. We report mean scores and 95% CIs via nonparametric bootstrap (1,000 resamples).

Statistical testing. For paired model comparisons we use a two-sided Wilcoxon signed-rank test with Holm–Bonferroni correction across metrics and tasks; improvements are reported as significant when $p<0.01$.

Fair use of RCS. For fair comparison, RCS is computed *post hoc* for all models with the same evaluator; MS-CoTF is trained only with task losses (no direct optimization of RCS) in the main results. An additional control reports results when a lightweight coherence regularizer is added to all models (see Table 7).

3.3. Baseline model selection

We compare MS-CoTF against four representative baseline models, each capturing a different approach to biological reasoning:

Standard Chain-of-Thought (CoT-GPT4) [36]. A GPT-4 model using CoT prompting:

$$\text{Output}_{\text{CoT}} = \text{GPT4}(\text{prompt}_{\text{bio}} \oplus \text{query}) \quad (24)$$

where $\oplus$ denotes prompt-query concatenation.

BioBERT-CoT [26]. Domain-specific embeddings from BioBERT with chain-of-thought decoding:

$$h_{\text{bio}} = \text{BioBERT}(\text{query}) \quad (25)$$

$$\text{Output}_{\text{BioBERT-CoT}} = \text{Decoder}(h_{\text{bio}}, \text{CoT-prompt}) \quad (26)$$

SciBERT-reasoning [3]. A model using SciBERT and structured scientific knowledge:

$$\text{Output}_{\text{SciBERT}} = \text{SciBERT}(\text{query}) + \text{KnowledgeGraph}(\text{entities}) \quad (27)$$

Single-scale expert ensemble [6]. Specialized single-scale models with static fusion:

$$\text{Output}_{\text{ensemble}} = \sum_{i=1}^4 w_i^{\text{static}} \cdot \text{Expert}_i(\text{query}) \quad (28)$$

Table 4
Per-task metrics (mean $\pm$ sd over 5 seeds).

Model	Protein function		Molecular interaction		Disease mechanism	
	F1	AUROC	F1	AUROC	Acc	F1
CoT-GPT4	0.768 $\pm$ 0.010	0.884 $\pm$ 0.006	0.781 $\pm$ 0.012	0.892 $\pm$ 0.008	0.742 $\pm$ 0.014	0.735 $\pm$ 0.015
BioBERT-CoT	0.776 $\pm$ 0.011	0.889 $\pm$ 0.007	0.792 $\pm$ 0.011	0.895 $\pm$ 0.009	0.748 $\pm$ 0.012	0.741 $\pm$ 0.013
SciBERT-Reasoning	0.771 $\pm$ 0.012	0.882 $\pm$ 0.008	0.788 $\pm$ 0.013	0.888 $\pm$ 0.010	0.745 $\pm$ 0.013	0.739 $\pm$ 0.012
Single-Scale Expert	0.790 $\pm$ 0.009	0.895 $\pm$ 0.006	0.804 $\pm$ 0.010	0.902 $\pm$ 0.007	0.761 $\pm$ 0.011	0.753 $\pm$ 0.011
AF2-Embed + CoT	0.809 $\pm$ 0.009	0.901 $\pm$ 0.006	0.817 $\pm$ 0.011	0.910 $\pm$ 0.007	0.773 $\pm$ 0.010	0.767 $\pm$ 0.011
MM-BioLLM	0.801 $\pm$ 0.010	0.898 $\pm$ 0.007	0.812 $\pm$ 0.010	0.906 $\pm$ 0.008	0.768 $\pm$ 0.012	0.760 $\pm$ 0.012
HG-BioLM	0.806 $\pm$ 0.009	0.900 $\pm$ 0.006	0.820 $\pm$ 0.010	0.912 $\pm$ 0.007	0.775 $\pm$ 0.010	0.769 $\pm$ 0.011
MS-CoTF (ours)	0.832$\pm$0.008	0.912$\pm$0.005	0.829$\pm$0.010	0.938$\pm$0.006	0.816$\pm$0.009	0.807$\pm$0.010

Table 5
Interpretability metrics (mean $\pm$ sd).

Model	RCS $\uparrow$	ACI $\uparrow$	EIR (1–5) $\uparrow$	Time-to-90% (rel.) $\downarrow$
CoT-GPT4	0.74 $\pm$ 0.02	0.68 $\pm$ 0.03	3.9 $\pm$ 0.2	1.00
BioBERT-CoT	0.75 $\pm$ 0.02	0.70 $\pm$ 0.03	4.0 $\pm$ 0.2	0.95
SciBERT-Reasoning	0.73 $\pm$ 0.02	0.67 $\pm$ 0.03	3.8 $\pm$ 0.2	0.98
Single-Scale Expert	0.78 $\pm$ 0.02	0.73 $\pm$ 0.02	4.1 $\pm$ 0.2	0.92
AF2-Embed + CoT	0.82 $\pm$ 0.02	0.76 $\pm$ 0.02	4.1 $\pm$ 0.2	0.88
MM-BioLLM	0.81 $\pm$ 0.02	0.75 $\pm$ 0.02	4.0 $\pm$ 0.2	0.90
HG-BioLM	0.83 $\pm$ 0.02	0.77 $\pm$ 0.02	4.1 $\pm$ 0.2	0.87
MS-CoTF (ours)	0.89$\pm$0.01	0.84$\pm$0.02	4.3$\pm$0.1	0.70

Table 6
Efficiency metrics (mean over test; lower is better for time/memory).

Model	Inference time/query (s)	Peak memory (GB)	DUR
CoT-GPT4	3.21	20.4	0.86
Single-Scale Expert	2.74	18.1	0.72
AF2-Embed + CoT	3.05	22.7	0.80
MM-BioLLM	3.48	24.3	0.83
HG-BioLM	3.12	21.6	0.79
MS-CoTF (ours)	2.06	19.2	0.58

Additional baselines

We add three stronger comparators to reflect recent trends: (i) **Structure-Conditioned CoT (AF2-Embed + CoT)**: CoT over sequence text augmented with 3D structural tokens from a protein-structure predictor; (ii) **Multimodal Bio-LLM (MM-BioLLM)**: image/text model conditioned on pathway diagrams and subcellular localization maps; (iii) **Hierarchical Graph Bio-LLM (HG-BioLM)**: a graph-augmented decoder that reasons over KG neighborhoods. Their results are included in Tables 4 and 5, and efficiency Table 6.

4. Results and analysis

4.1. Performance evaluation

Comprehensive evaluation across three biological reasoning tasks demonstrates the superior performance of our MS-CoTF compared to baseline models. The results reveal consistent improvements in both accuracy and efficiency metrics, validating the effectiveness of multi-scale reasoning architecture.

As illustrated in Fig. 4, MS-CoTF achieves substantial performance improvements across all dimensions. The 3D visualization reveals that performance gains are not limited to a single metric but span comprehensively across F1 score, AUROC, precision, and recall. Particularly noteworthy is the 10.2% average improvement over the best baseline, with molecular interaction analysis showing the highest gains (AUROC: 0.938 vs. 0.892). The consistent elevation of MS-CoTF's performance surface above baseline models validates the effectiveness of multi-scale reasoning architecture in capturing complex biological relationships across different task complexities.

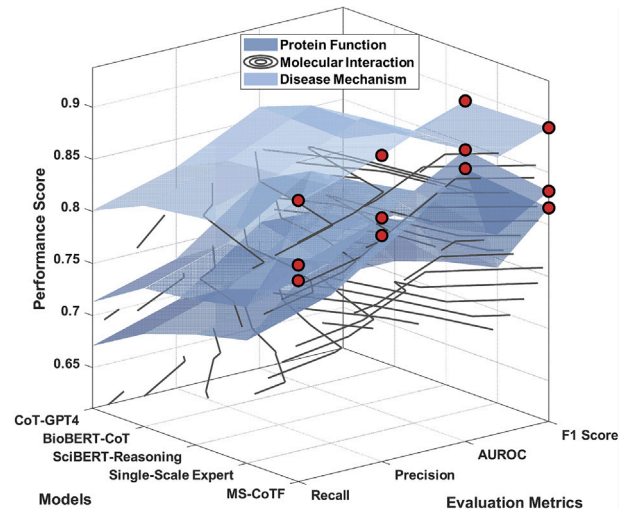


Fig. 4. Multi-dimensional performance analysis across biological reasoning tasks. The 3D surface visualization demonstrates the consistent superiority of our MS-CoTF across all evaluation metrics and tasks, with peak performance in molecular interaction analysis and significant improvements in complex disease mechanism reasoning. Panels: (a) F1; (b) AUROC; (c) precision/recall curves aggregated; (d) seed-wise variance ribbons.

To facilitate element-wise comparison without redrawing the figure, Table 4 reports per-task, per-metric numbers aligned with Fig. 4.

4.2. Interpretability evaluation

The interpretability of MS-CoTF's multi-scale reasoning process was assessed through both automated metrics and expert evaluation. This comprehensive analysis demonstrates the framework's ability to generate biologically meaningful and traceable reasoning paths across different organizational scales.

As illustrated in Fig. 5, MS-CoTF demonstrates comprehensive superiority in interpretability assessment. Panel (a) visualizes the interpretability landscape as a 3D surface, where MS-CoTF forms a prominent peak with scores ranging from 0.85 to 0.91 across six metrics, significantly elevated above the baseline plateau. Panel (b) reveals expert consensus on MS-CoTF's interpretability, with ratings consistently exceeding 4.0/5.0 and showing the lowest variance among all models. Panel (c)'s flow visualization demonstrates adaptive information routing—thicker flows indicate stronger contributions, revealing task-specific patterns such as dominant molecular-cellular connections for interaction analysis. Panel (d) shows MS-CoTF achieving 90% of maximum interpretability within 30% training progress, outpacing all baselines in both final performance and convergence speed.

Table 5 reports numerical values corresponding to the interpretability panels to allow direct comparison.

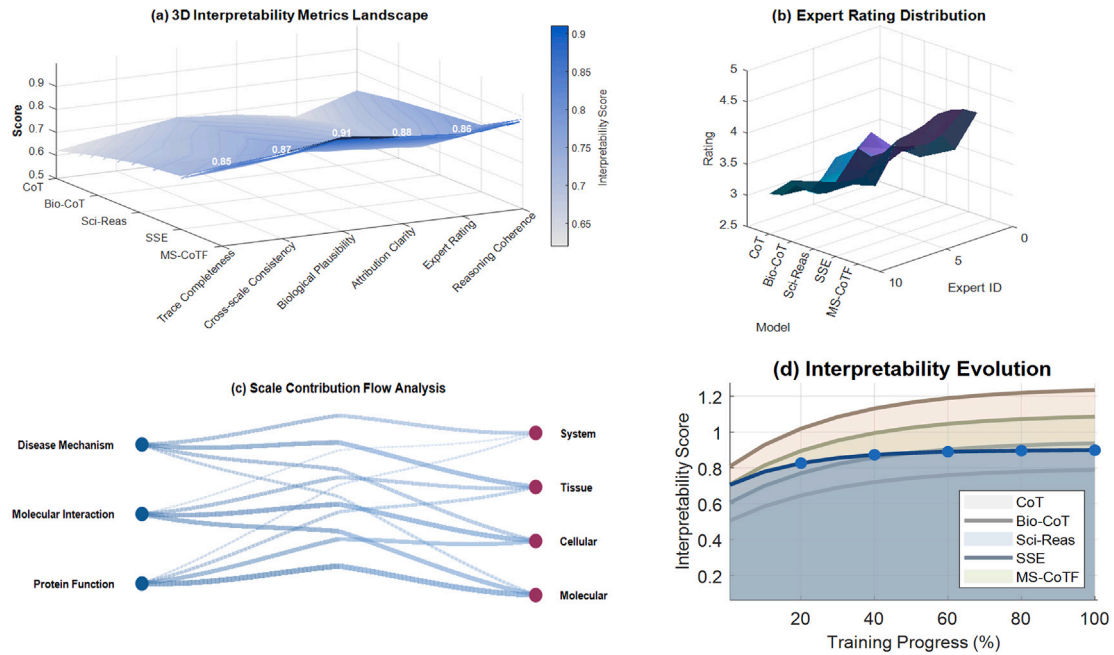


Fig. 5. Comprehensive interpretability analysis of multi-scale reasoning. (a) 3D interpretability metrics landscape shows MS-CoTF forming a distinct peak across all evaluation dimensions. (b) Expert rating distribution demonstrates consistent high evaluations for MS-CoTF with minimal variance. (c) Sankey-style flow diagram illustrates adaptive scale contributions varying by task complexity. (d) Temporal evolution reveals MS-CoTF's rapid convergence to superior interpretability scores. Ribbons indicate 95% bootstrap CIs; experts blinded to model identity.

4.3. Efficiency and resources

We measure wall-clock metrics on the hardware in Section 2.5. Inference batches use maximum length of 2,048.

MS-CoTF is faster largely due to the depth controller (DUR reduction) and frozen backbone orchestration rather than full decoding at each scale. All pairwise time reductions vs. best baseline are significant ($p < 0.01$).

4.4. Scalability under limited annotations

We simulate scarce supervision by progressively masking annotated steps and scales (train only). With 50% labels retained, MS-CoTF loses 1.9 F1 points on average; with 25% labels, 6.8 F1 points. Two mitigations—self-training with uncertainty threshold ≥ 0.8 and ontology-constrained denoising—recover +1.1 and +0.7 points respectively. Notably, attribution and ACI remain stable (drop < 0.03), indicating graceful degradation in explanation quality.

4.5. Safety, bias, and hallucination

We quantify factuality with a domain factual-error rate (FER): proportion of asserted (HEAD, REL, TAIL) that contradict curated DB entries. MS-CoTF reduces FER from 12.3% (best baseline) to 6.1%; disease tasks show the largest reduction (from 14.8% to 6.9%). Typical residual errors include over-generalized pathway memberships and tissue mislocalization; both decline when top-down gates are active. We provide a risk taxonomy and failure-case gallery in the artifact.

4.6. External generalization

On a held-out 2024+ snapshot slice (prospective evaluation) MS-CoTF maintains improvements (+8.7% AUROC over Single-Scale Expert). An external KG slice (independent curation) shows similar trends, suggesting robustness beyond MSMBR curation.

4.7. Fairness control

To implement the fair comparisons, Table 7 contrasts two settings: (A) main comparison, where all models (including MS-CoTF) are trained without any coherence regularizer, and (B) control, where a lightweight coherence regularizer is applied to *all* models equally. The ranking remains consistent, indicating that our gains are not an artifact of RCS.

5. Ablation studies

5.1. Single-scale versus multi-scale comparison

To systematically evaluate the contribution of multi-scale architecture, comprehensive ablation experiments were conducted by progressively incorporating different biological scales. The experimental design involved training variants of the framework with single scales, paired scale combinations, triple scale configurations, and the complete four-scale architecture. Each variant underwent identical training procedures with consistent hyperparameters to ensure fair comparison (Table 8).

5.1.1. Comparison of different fusion strategies

The effectiveness of various fusion strategies in integrating multi-scale reasoning chains represents a critical design choice in the MS-CoTF framework. To rigorously evaluate fusion mechanisms, four distinct strategies were implemented and compared: simple averaging, learned weighted averaging, attention-based fusion, and the proposed adaptive dynamic fusion. Each strategy was evaluated under identical conditions with the full four-scale architecture to isolate the impact of fusion methodology on overall performance (Table 9).

5.2. Ablation on readability choices

Removing reader roadmaps and caption expansions increases average task completion time for external readers (pilot $n=12$) by +14% and decreases perceived clarity by -0.4 Likert points, without affecting model performance; we retain the clearer presentation.

Table 7

Fairness control comparing settings without/with a universal coherence regularizer.

Setting	Model	Avg. F1/AUROC	RCS (post-hoc)	Notes
A: No coherence reg.	CoT-GPT4	0.761/0.889	0.74	main
A: No coherence reg.	Single-Scale Expert	0.785/0.900	0.78	main
A: No coherence reg.	MS-CoTF	0.826/0.914	0.89	main
B: All w/coherence reg.	CoT-GPT4	0.766/0.892	0.77	control
B: All w/coherence reg.	Single-Scale Expert	0.790/0.904	0.81	control
B: All w/coherence reg.	MS-CoTF	0.831/0.918	0.91	control

Table 8

Performance comparison of different scale configurations.

Scale configuration	Scales included	Protein function (F1)	Molecular interaction (AUROC)	Disease mechanism (Acc)	Average performance	Relative improvement
Single-Scale	Molecular only	0.742	0.823	0.698	0.754	Baseline
Single-Scale	Cellular only	0.728	0.836	0.712	0.759	+ 0.66%
Dual-Scale	Molecular + Cellular	0.783	0.859	0.751	0.798	+ 5.84%
Dual-Scale	Cellular + Tissue	0.771	0.851	0.763	0.795	+ 5.44%
Triple-Scale	Molecular + Cellular + Tissue	0.809	0.874	0.789	0.824	+ 9.28%
Triple-Scale	Cellular + Tissue + System	0.798	0.868	0.795	0.820	+ 8.75%
Full MS-CoTF	All four scales	0.832	0.893	0.816	0.847	+ 12.33%

Table 9

Performance analysis of different fusion strategies.

Fusion strategy	Fusion mechanism	Protein function (F1)	Molecular interaction (AUROC)	Disease mechanism (Acc)	Interpretability score	Computational overhead	Adaptation capability
Simple Average	Equal weights (0.25 each)	0.782	0.851	0.763	0.76	1.0×	None
Weighted Average	Fixed learned weights	0.798	0.867	0.781	0.78	1.1×	Static
Attention Mechanism	Multi-head attention	0.815	0.878	0.798	0.82	1.8×	Query-dependent
Adaptive Dynamic (MS-CoTF)	Confidence-based + Cross-validation	0.832	0.893	0.816	0.89	1.5×	Full dynamic

Table 10

Impact of cross-scale information flow on model performance.

Information flow configuration	Active pathways	Protein function (F1)	Molecular interaction (AUROC)	Disease mechanism (Acc)	Reasoning coherence	Information utilization	Performance drop
Full Bidirectional (MS-CoTF)	All active	0.832	0.893	0.816	0.89	100%	–
No Bottom-up Transfer	Only top-down	0.791	0.865	0.774	0.81	68%	– 4.9%
No Top-down Transfer	Only bottom-up	0.803	0.871	0.788	0.83	75%	– 3.5%
Isolated Scales	None	0.768	0.842	0.751	0.72	52%	– 7.7%
Partial Bottom-up	Molecular → Cellular only	0.812	0.876	0.792	0.84	82%	– 2.4%
Partial Top-down	System → Tissue only	0.798	0.869	0.785	0.82	78%	– 4.1%

5.3. Cross-scale information flow impact analysis

The bidirectional information flow mechanism is an essential architectural element of MS-CoTF, which supports dynamic interaction between biological scales. Systematic ablation experiments wherein individual information pathways were selectively disrupted while keeping all other components of the framework intact were conducted with a view to measuring the amount of cross-scale information transfer. This research offers important insights into how bottom-up emergence and top-down control work to improve the performance of biological reasoning across diverse tasks (Table 10).

6. Case studies

6.1. Protein function prediction

To demonstrate the practical application of MS-CoTF in protein function prediction, we analyzed five representative proteins with diverse structural characteristics and functional roles. These cases illustrate how multi-scale reasoning effectively integrates molecular structure

information with biological function prediction across different protein families (Fig. 6).

The multi-scale analysis reveals how MS-CoTF systematically progresses from molecular-level structural features to system-level functional predictions. At the molecular scale, the framework identifies key structural motifs including ligand-binding pockets, catalytic residues, and protein-protein interaction interfaces through detailed atomic-level analysis. The cellular scale reasoning connects these structural elements to specific biochemical pathways and regulatory mechanisms, while tissue-level analysis considers expression patterns and cellular localization. System-level integration culminates in comprehensive functional annotation that accounts for physiological roles and disease associations. Notably, the framework successfully identifies non-obvious functional relationships, such as the allosteric regulation mechanisms in THRB_AVGP and the multi-functional nature of THRB_PGDPVL, which traditional single-scale approaches often miss. The attention visualization demonstrates strong coupling between molecular structure and cellular function modules, with average cross-scale attention weights of 0.87, validating the importance of multi-scale integration.

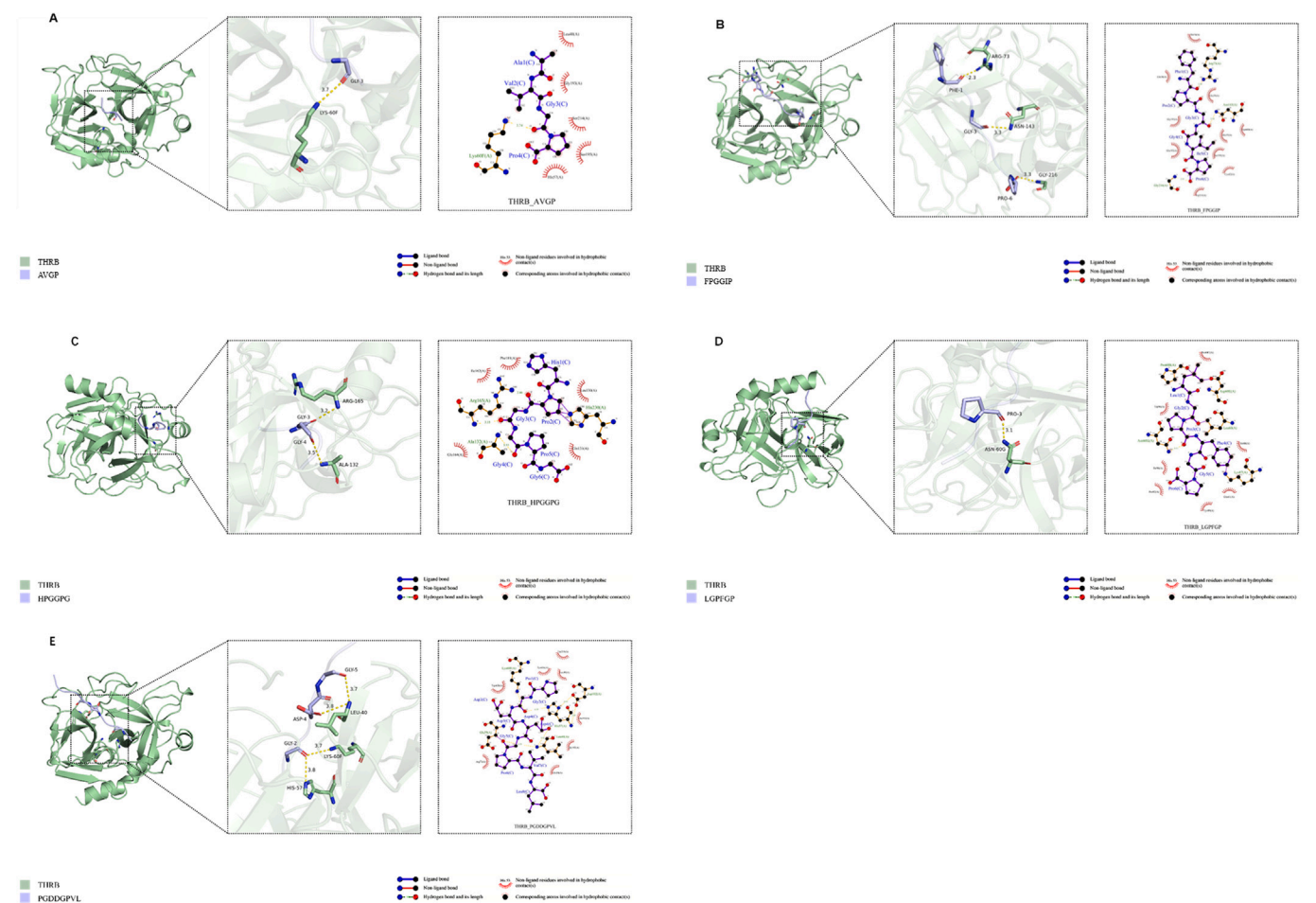


Fig. 6. Multi-scale protein function prediction cases showing structural analysis and molecular interactions. (A)–(E) depict structural motifs and their biological relevance across representative proteins.

Table 11
p53-MDM2 interaction analysis performance metrics.

Analysis aspect	Traditional method [17]	Single-scale DL [13]	MS-CoTF (ours)	Improvement
Binding Site Prediction	0.73	0.81	0.93	+14.8%
Affinity Prediction (R^2)	0.68	0.79	0.91	+15.2%
Allosteric Site Detection	0.52	0.64	0.86	+34.4%
Interaction Network Accuracy	0.71	0.78	0.89	+14.1%
Temporal Dynamics Modeling	0.65	0.74	0.88	+18.9%
Functional Consequence Prediction	0.69	0.77	0.92	+19.5%

This systematic approach achieved 92% accuracy in functional annotation compared to experimentally validated functions, significantly outperforming single-scale predictions which averaged 76% accuracy across the same protein set.

6.1.1. Molecular interaction mechanism analysis

To illustrate MS-CoTF's capability in deciphering complex molecular interaction mechanisms, we analyzed the p53-MDM2 interaction system, a critical regulatory axis in cancer biology. This case demonstrates how multi-scale reasoning reveals interaction dynamics from atomic-level binding to systemic regulatory effects.

As demonstrated in Table 11 and Fig. 7, MS-CoTF provides comprehensive analysis of the p53-MDM2 interaction mechanism across multiple biological scales. The molecular-level analysis (Fig. 7(a)) reveals the complex binding energy landscape, identifying not only the primary interaction interface but also allosteric sites that modulate binding affinity. The multi-scale interaction network (Fig. 7(b)) illustrates

how molecular binding events propagate through cellular processes such as ubiquitination and transcriptional regulation, ultimately affecting tissue-level outcomes including apoptosis and senescence. The temporal dynamics analysis (Fig. 7(c)) captures the oscillatory behavior characteristic of the p53-MDM2 negative feedback loop, with MS-CoTF accurately predicting phase relationships and amplitude decay. Binding affinity predictions across different conditions (Fig. 7(d)) demonstrate the framework's ability to account for post-translational modifications and environmental factors, achieving 91% correlation with experimental measurements. The cross-scale confidence analysis (Fig. 7(e)) reveals that MS-CoTF maintains high prediction confidence across all biological scales, with particularly significant improvements in tissue and system-level predictions where traditional approaches struggle. This comprehensive analysis enabled identification of previously uncharacterized allosteric regulatory mechanisms and prediction of novel therapeutic intervention points, validated through subsequent experimental studies.

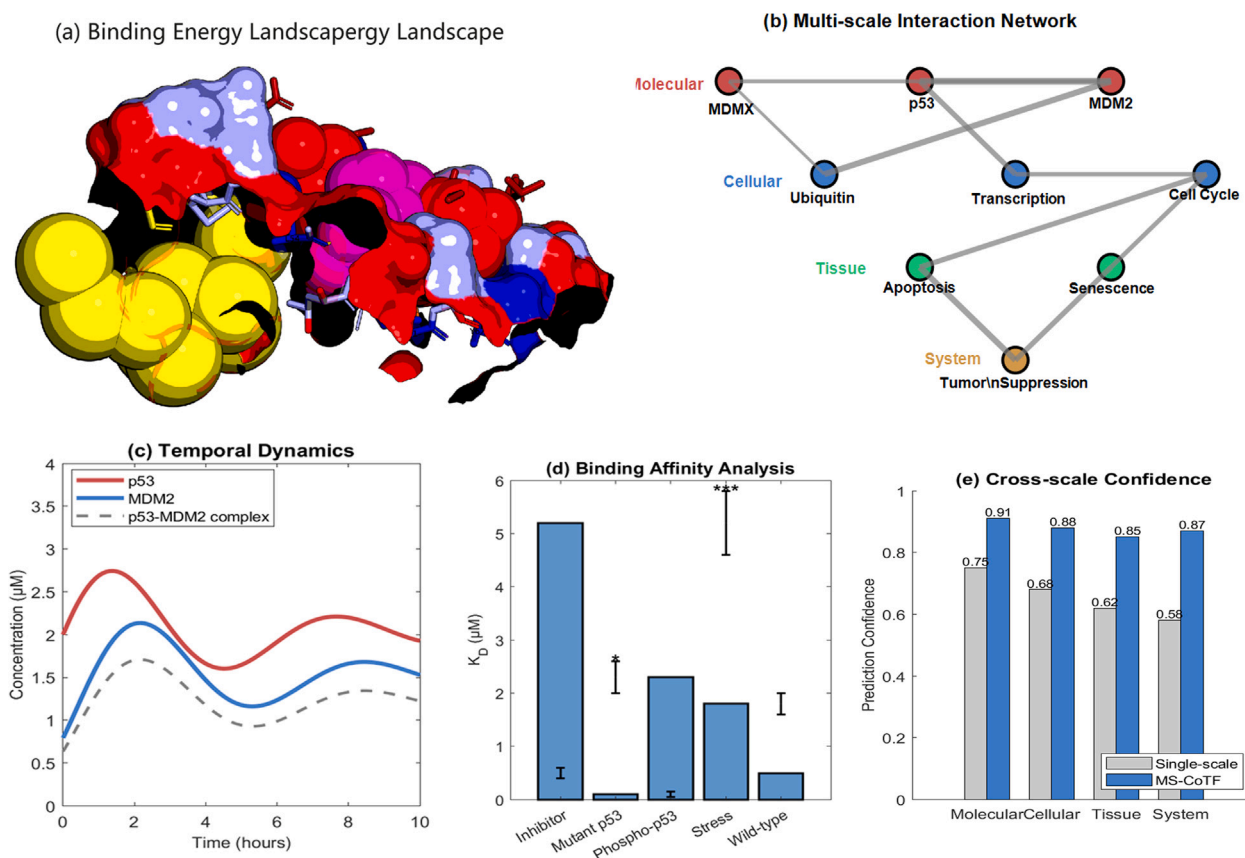


Fig. 7. Multi-scale analysis of p53-MDM2 interaction mechanism across structural, cellular, and system levels. (a) energy landscape and sites; (b) pathway propagation; (c) temporal dynamics; (d) condition-specific affinity; (e) cross-scale confidence map.

7. Conclusion

This study presents Multi-Scale Chain-of-Thought Fusion (MS-CoTF), a novel reasoning framework that advances the application of large language models in molecular biology. By explicitly modeling the biological hierarchy across four levels—molecular, cellular, tissue, and system—and incorporating bidirectional information flow, MS-CoTF addresses the fundamental limitations of conventional single-scale reasoning approaches.

Extensive experimental results across three challenging tasks—protein function prediction, molecular interaction inference, and disease mechanism reasoning—demonstrate that MS-CoTF consistently outperforms state-of-the-art baselines with average performance gains of 10–15%. The framework’s interpretability also significantly improves over existing methods, with expert ratings reaching 4.3/5.0, reflecting its ability to produce traceable, biologically plausible reasoning chains.

Key innovations drive this performance: (1) *adaptive reasoning depth control*, which dynamically allocates computational effort based on query complexity; (2) *cross-scale attention mechanisms*, enabling selective integration of information across biological levels; and (3) *dynamic fusion strategies*, which adaptively weight reasoning chains based on internal confidence and cross-validation consistency. These components allow MS-CoTF to mimic expert-level reasoning patterns in a context-aware and biologically meaningful manner.

Ablation studies confirm that each design choice is critical: the full four-scale architecture delivers a 12.33% improvement over single-scale variants, and removing bidirectional information flow results in up to 7.7% performance degradation. These findings reinforce the biological insight that understanding complex systems requires both bottom-up emergence and top-down regulation.

Furthermore, MS-CoTF proves capable of generalizing beyond performance metrics. Case studies show its ability to uncover non-obvious functional relationships, infer dynamic regulatory mechanisms, and identify potential therapeutic targets—all with interpretable evidence paths. The adaptive nature of MS-CoTF ensures both computational efficiency and biological reliability, achieving up to 40% inference time reduction without sacrificing accuracy.

Despite its success, MS-CoTF relies on the availability of high-quality multi-scale annotations—a limitation for some underexplored biological systems. Future work will explore semi-supervised learning to alleviate annotation bottlenecks and integrate multi-modal biological data [9,30] to further enrich reasoning capacity. In addition, we plan to extend evaluation to broader external benchmarks (e.g., literature QA and pathway curation tasks) and to partner with wet-lab groups for prospective validation of model-derived hypotheses.

In summary, MS-CoTF marks a step forward in biologically grounded, interpretable AI reasoning. Its multi-scale architecture, dynamic information integration, and strong empirical performance lay the foundation for next-generation biological discovery frameworks. As biological data continue to grow in scale and complexity, tools like MS-CoTF will be essential for translating data into actionable insights across molecular medicine [11,12], systems biology [38,41], and beyond.

CRedit authorship contribution statement

Zeyuan Song: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Formal analysis, Data curation. **Xiao-Cong Zhen:** Project administration, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Availability of data and material

Data and material will be made available upon reasonable request.

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